Data Analytics and Machine Learning for solar power generation

ENERGY DATA ANALYTICS

Paolo Finardi

Energy & Finance Institute

www.energyandfinanceinstitute.com

Paolo Finardi

Head of
Data Science & AI

- **Leads AI Strategy** and a team of 30+ Data Scientists across the Retail, E-Mobility, and Renewables sectors.
- **10+ years of experience** focused on scalable ML solutions in Forecasting and Energy Optimization.
- **Active community contributor**: Member of the Databricks Product Advisory Board and Organizer of PyData Milan.

www.linkedin.com/in/paolo-finardi-38b25096

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Transition to data-driven organizations

Energy companies are evolving by integrating data science to improve operations and decision-making

Data Science Benefits

Data science turns raw data into insights for better decisions.

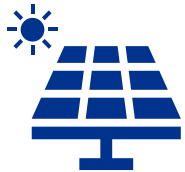
Examples of data science project on PV industry

Predict the next day's PV production.

Identify inefficiencies in a plant.

Optimize storage usage.

Examples of data science project on PV industry



Predict the next day's PV production.

Data sources

- SCADA: active power (P), inverter status, temperature, DC/AC power
- Weather forecasts: irradiance, temperature, wind speed, cloud cover
- Historical production logs
- Asset metadata: location, tilt, azimuth, installed capacity

Model Class

- Time Series & Forecasting Models:

Statistical → ARIMA, SARIMA, ETS.
Machine Learning → XGBoost, LSTM, Random Forest
Decomposable models → Fb Prophet

Examples of data science project on PV industry

Identify inefficiencies in a plant.

<u>Data sources</u>

- SCADA + sensors: DC power, AC output, inverter temperatures.
- Weather stations: irradiance, ambient & module temperature, wind.
- Asset design parameters: panel efficiency, inverter curve, degradation factors.
- Maintenance logs / alarms.

<u>Model Class</u>

- Hybrid or Physics-informed Regression Models:

Empirical models -> Performance Ratio (PR), custom regression of expected yield vs irradiance.

Data-driven anomaly detection -> Isolation Forest, Autoencoder, One-Class SVM.

Hybrid approaches combining physics-based expected output with ML residual analysis.

Examples of data science project on PV industry

Optimize storage usage.

<u>Data sources</u>

- Production and forecast data (from Example 1).
- Asset health indicators and maintenance logs.
- Energy prices, market tariffs, and constraints
- Battery SoC, charge/discharge cycles, and inverter efficiency.

<u>Model Class</u>

- Optimization / Decision Models:

Linear or Mixed-Integer Programming (MILP).
Constraint optimization (CP-SAT, Gurobi, Pyomo).
Reinforcement Learning (for adaptive control or storage dispatch).

Data Scienct Project Lifecycle

Business Understanding

Define project objectives and understand business requirements to guide data science efforts.

Data Understanding

Collect and explore data to identify quality, patterns, and initial insights.

Data Preparation

Clean and transform data to create a suitable dataset for modeling and analysis.

Modelling

Build models using algorithms to extract patterns and make predictions.

Evaluation

Assess model performance using metrics to ensure accuracy and reliability.

Deployment & Business Translation

Implement the model in production and translate results into actionable business insights.

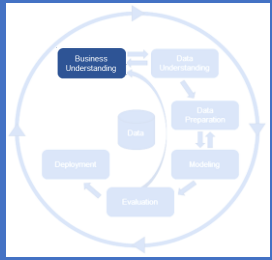
Iterative nature and deliverables of each phase

Phase-Specific Questions
Each project phase addresses a unique question guiding the next steps and decision-making.

Tangible Deliverables
Every phase produces clear, tangible deliverables that provide milestones and progress validation.

Iterative Process Flow
Projects iterate by moving back and forth between phases → continuous refinement.

1) Business Understanding



Phase-Specific Questions
What problem are we solving?
What metric defines success?

Tangible Deliverables
Business Canvas

Start with Business Questions

Begin by framing clear business questions that focus on measurable impact and actionable outcomes.

Define Success Metrics

Set clear success metrics such as MAPE targets, cost savings, and environmental impact for effective evaluation.

Identify Stakeholders and Timeline

Recognize key stakeholders, decisions affected, and the time horizon for implementing solutions.

PRODUCT

BUSINESS

Problem

- What is the problem?
- Why is it a problem?
- Whose problem is it?

1.

Data

- Source
- Quality
- Availability
- Input / Output

3.

Hypothesis

- What hypotheses to test?
- Expected outcomes?
- How to accommodate the results?

4.

Solution

- Type of the solution?
- What will be the solution?
- Outcome expected?

2.

KPIs

- How to evaluate the model?
- What metrics to use?

7.

Actors

- Who is the client, stakeholder, and sponsor?
- Who will be impacted by the solution?

5.

Actions

- What are the actions to be taken?

6.

Values

8.

- What is the extent of the problem?
- What is the baseline?
- What are the expected assets?

Risks

9.

- What are the risks?
- What do they impact?

Performance / Impact

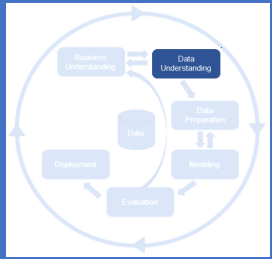
- What is the business impact?
- Where and how it can be seen?
- How to measure it?

10.

STRATEGY

https://www.acronymat.com/2023/08/22/data-model-canvas/

2) Data Understanding



Phase-Specific Questions
What data do we have?
Is it reliable?

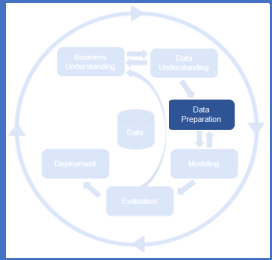
Tangible Deliverables
Data inventory,
initial exploration

Data Collection & Inventory
Identify all available data sources and their structure (type, frequency, format, variables).

Data Quality Assessment
Evaluate completeness, consistency, and accuracy of the datasets.

Initial Exploration
Perform preliminary analyses to uncover distributions, correlations, and potential anomalies.

3) Data Preparation



Phase-Specific Questions
How do we clean and structure the dataset?

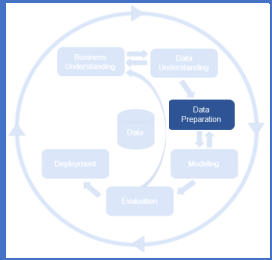
Tangible Deliverables
Ready-to-model dataset (Master Table)

Handling Missing Values
Identify and manage missing data to ensure dataset completeness and accuracy for modeling.

Correcting Outliers
Detect and adjust outliers to improve data quality and model reliability.

Feature Set Creation
Relevant features are extracted from training data to create the final feature set for modeling.

3) Data Preparation



Phase-Specific Questions
How do we clean and structure the dataset?

Tangible Deliverables
Ready-to-model dataset (Master Table)

Feature Set Creation

Lag Features
Lag features use past time points like P t-1 and P t-24 to capture temporal dependencies in data.

Rolling Statistics
Rolling means or standard deviations smooth data to reveal trends and variability over time.

Calendar and Weather Features
Features include hour, weekday, season and weather indices like clear-sky irradiance for richer input data.

Cyclic and Interaction Features
Cyclic encodings like sin/cos for time and interactions between variables enhance feature informativeness.

Hands-on

Data Understanding & Data Preparation

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Hands-On Session: Solar Power Generation Analysis

Objective

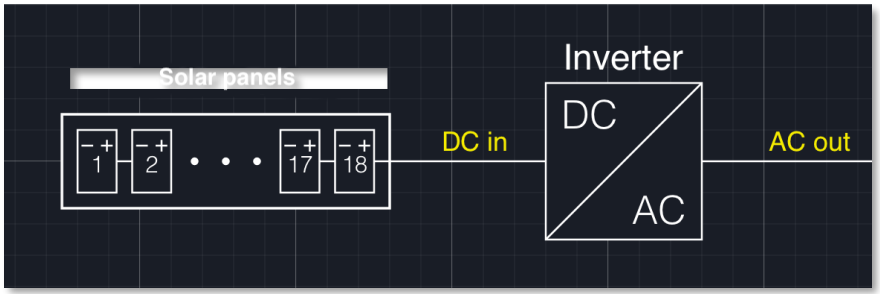
Apply the full analytics workflow on a real-world photovoltaic dataset to explore, clean, and model solar power production.

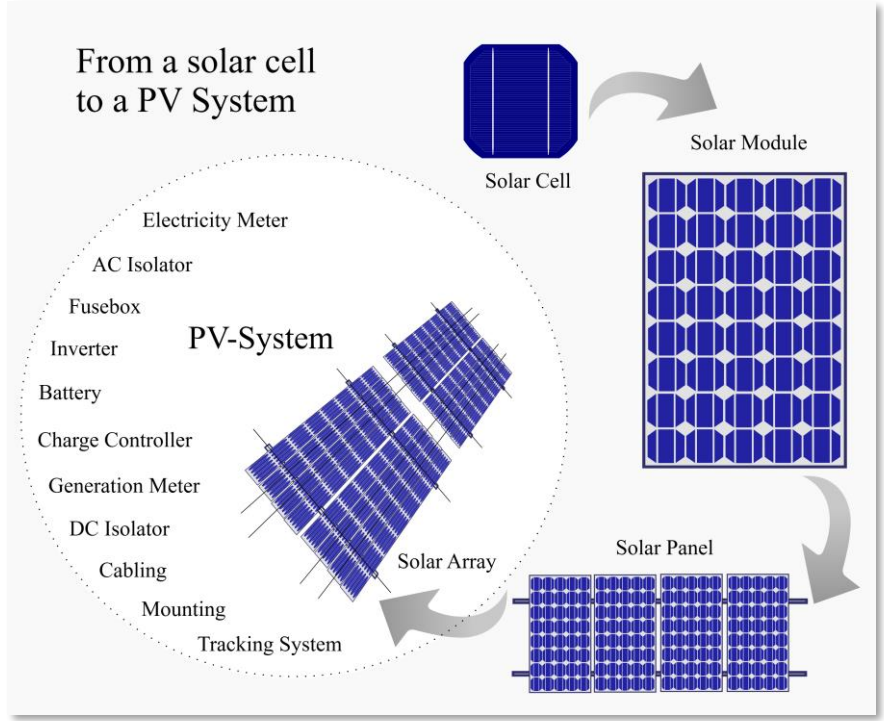
Dataset

Kaggle – Solar Power Generation Data

- 34 days of generation data from multiple plants and inverters
- Key features: DATE_TIME, PLANT_ID, SOURCE_KEY, DC_POWER, AC_POWER, DAILY_YIELD, TOTAL_YIELD, IRRADIATION
- Frequency: 15-minute intervals

Glossary


Solar panels
1
2
17
18
DC in
Inverter
DC
AC
AC out


From a solar cell
to a PV System
Solar Cell
Solar Module
Electricity Meter
AC Isolator
Fusebox
Inverter
Battery
Charge Controller
Generation Meter
DC Isolator
Cabling
Mounting
Tracking System
PV-System
Solar Array
Solar Panel

Notebooks links

Github Repo

01_data_exploration_plant1.ipynb

02_data_exploration_plant2.ipynb

03_data_exploration_plant2_singles.ipynb

04_data_modelling_plant2_forecasting.ipynb

Q & A

4) Data Modelling

Phase-Specific Questions
Which algorithms or approaches?

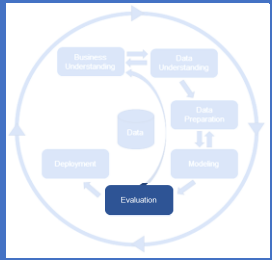
Tangible Deliverables
Predictive models object

Start from simple models
Simple baseline models, like persistence, provide foundational benchmarks for forecasting tasks.

Importance of Baseline Comparison
Compare new methods to a well-established baseline to ensure meaningful improvements.

Use Experiment books
Track models improvements with an experiment book to avoid rolling dice modeling

5) Evaluation



Phase-Specific Questions
Performance match the expectation? Is it stable?

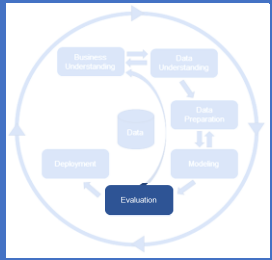
Tangible Deliverables
Metrics, validation results

Validation Techniques
Apply backtesting or rolling window validation to respect time order and simulate real operations.

Business-Relevant Metrics
Use MAE, RMSE, and MAPE for forecast depending on the use case

Linking Metrics to Business Impact
Explain how RMSE values translate to financial costs, highlighting operational relevance of metrics.

5) Evaluation



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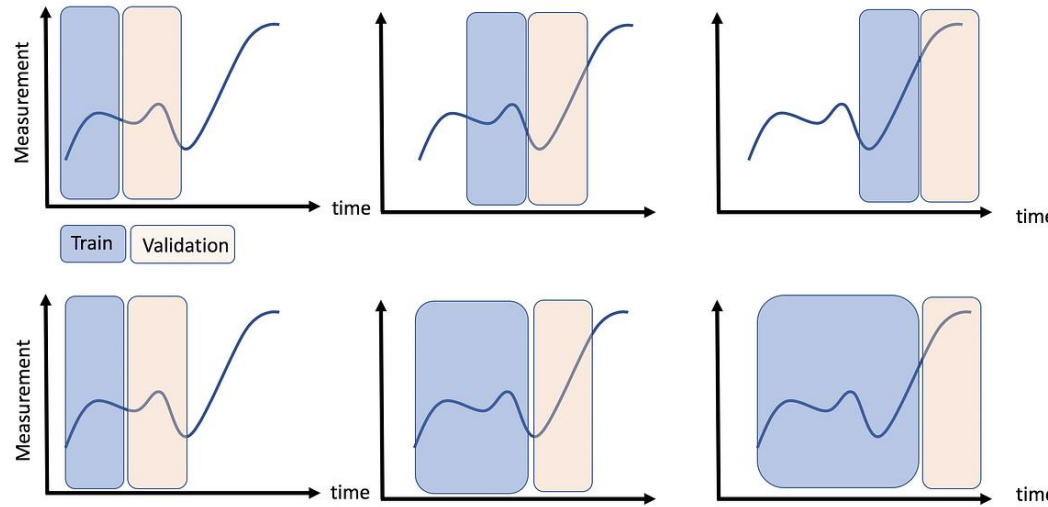


5) Evaluation

Evaluation Metrics for time series

Metric	Formula	What it Measures	Pros	Cons
MAE (Mean Absolute Error)	$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	Average magnitude of errors between predictions and actuals, without considering direction.	• Easy to interpret (same units as target). • Robust to outliers compared to RMSE.	- Treats all errors equally, doesn't penalize large deviations.
RMSE (Root Mean Squared Error)	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	Square root of the average of squared differences between predicted and actual values.	• Highlights large errors, good when big mistakes are costly. • Easy to interpret (same units as target).	• Sensitive to outliers, can overemphasize few large errors.
MAPE (Mean Absolute Percentage Error)	$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$	Average absolute percentage difference between predicted and actual values.	• Scale-independent, allows comparison across sites or systems. • Easy to communicate as percentage accuracy.	• Undefined when true values are zero

Iterative nature and deliverables of each phase

Phase	Key Questions	Deliverables
1. Business Understanding	What problem are we solving? What metric defines success?	Business Canvas
2. Data Understanding	What data do we have? Is it reliable?	Data inventory, initial exploration
3. Data Preparation	How do we clean and structure it?	Ready-to-model dataset
4. Modelling	Which algorithms or approaches?	Predictive models
5. Evaluation	Does it work? Is it usable?	Metrics, validation results
6. Deployment / Translation	How do we deliver the value?	Dashboard, decision support, automation

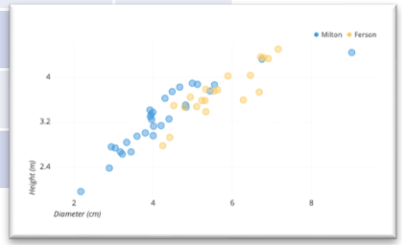
What is a time series

Cross-sectional data: observations at a fixed point in time or where time does not matter.

Time series: Datasets sorted and observed at successive points in time

Independent observations

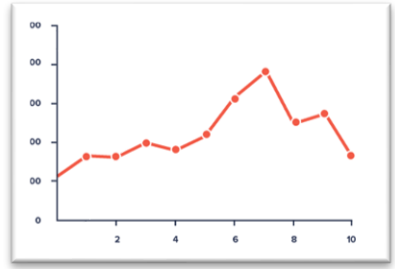
Indice	Y	X1	X2



Cross-sectional

Time-dependent

Time	Y



Time series

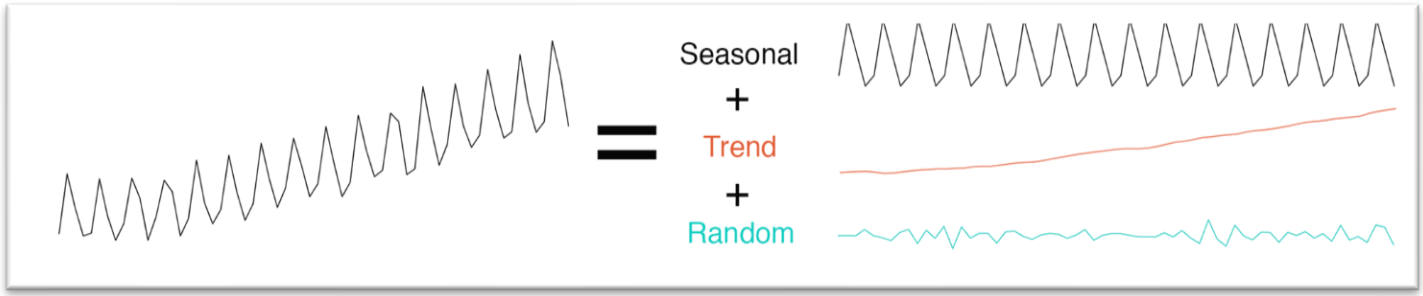
Time series decomposition

To make time series forecasts the simplest way is **decomposition**
In fact, each time series can be seen as the sum of a signal plus a noise

$$Time\ series\ =\ Signal\ +\ Noise$$

In the same way, the signal can be decomposed into different components such as trend and seasonality

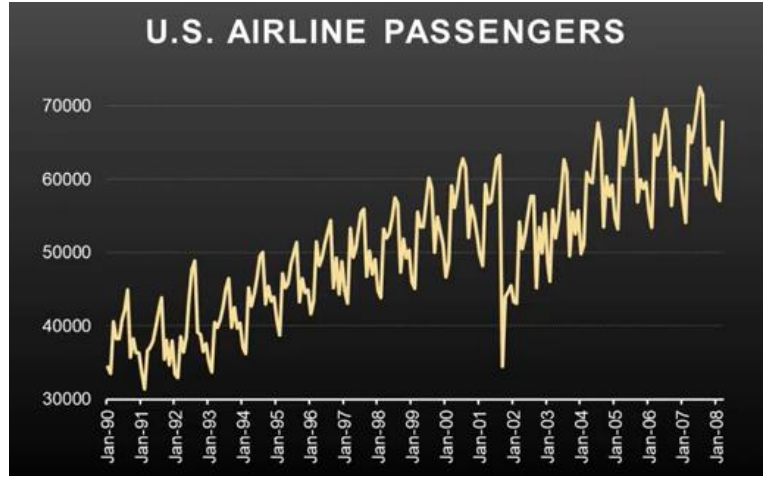
- Multiplicative models
  $Signal\ =\ trend\ *\ stagionalità * noise$
- Additive Models
  $Signal\ =\ trend\ +\ stagionalità + noise$



Classic component estimation

Additive Model

1. Calculate the trend g(t) on the series
2. Return the detrended series $y(t) - g(t)$
3. Calculate the seasonal contributions as averages of the detrended series for each of the m periods: we obtain $s(t)$.
4. The residuals are calculated as $y(t) - g(t) - s(t)$



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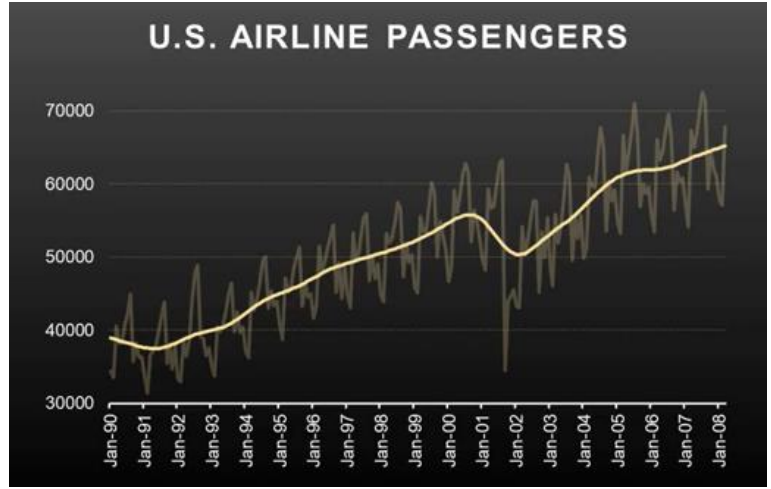

The easiest technique to calculate the trend is to make the **moving average**
There are more advanced techniques such as **LOESS Regression** – Locally estimated scatterplot smoothing

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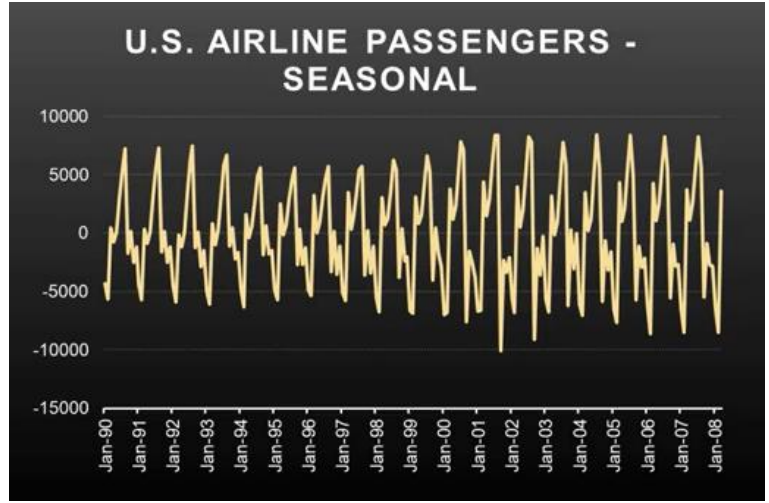

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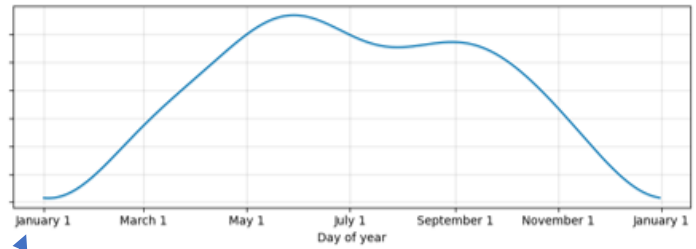

For example, to calculate the January contribution,
from the series adjusted for the average calculation trend of all January

Classic component estimation

Additive Model

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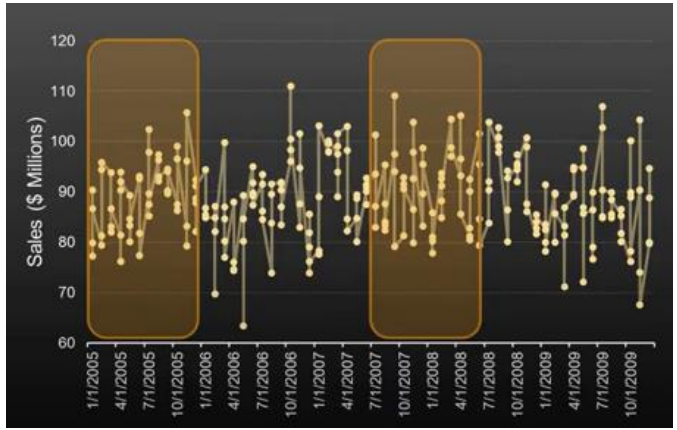
What is not predicted by the sum of trend + seasonality is called error

Stationary series



A series is said to be stationary if within **two intervals** of the **same size** we find a distribution with the **same mean** and the **same variance**

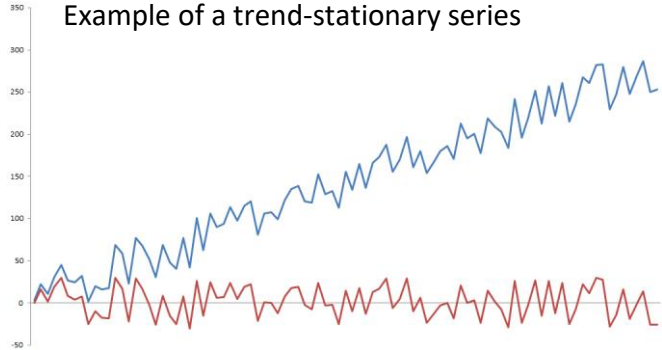
It is a necessary condition for estimating the components well!

Stationary series

A series can become stationary if

- Differentiated
- Trend-adjusted

Example of a trend-stationary series



One of the ways to identify non-stationary time series is to graph the autocorrelation function (ACF). For a stationary time series, the autocorrelation function will tend to zero rather quickly, while the autocorrelation function of non-stationary data decreases slowly

Facebook's Prophet

"Forecasting at Scale" di Taylor e Letham (2017)

Generalized Additive Model (GAM) with the aim of encouraging the use of domain knowledge within the forecasting model

Additive Model:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

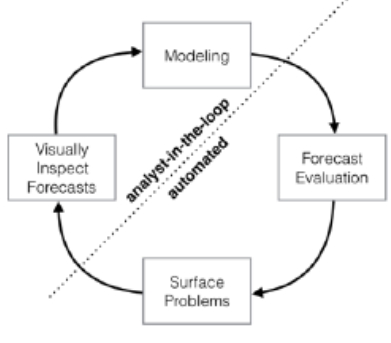
Regression on **trend** g(t), **seasonality** s(t), **special events** h(t)

Sum of functions that depend on time or time-dependent variables for whom I know the realization in the future:

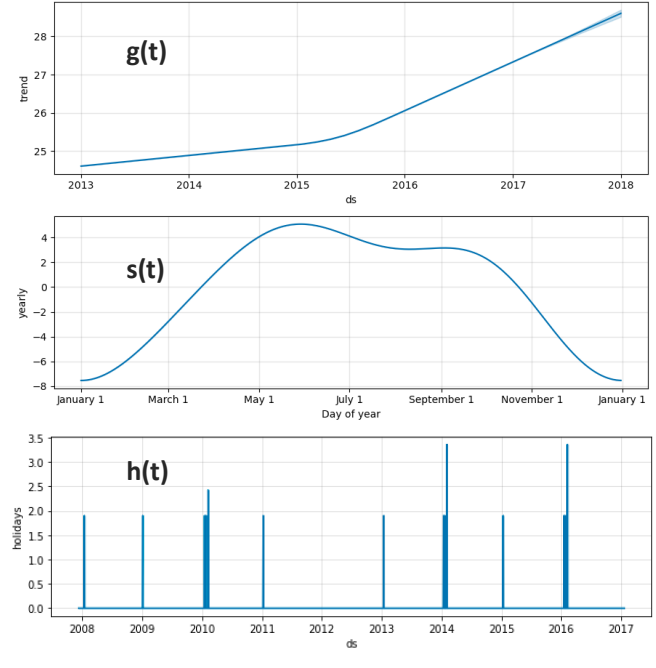
-> Forecast for t>t0

-> Express nonlinear dependencies

-> Resistant to outliers and missing values



Prophet's output



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Prophet Trend

Trend

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Linear with changepoints

$$g(t) = (k + \mathbf{a}(t)^\intercal \boldsymbol{\delta})t + (m + \mathbf{a}(t)^\intercal \boldsymbol{\gamma}),$$

k = base growth rate (cost)

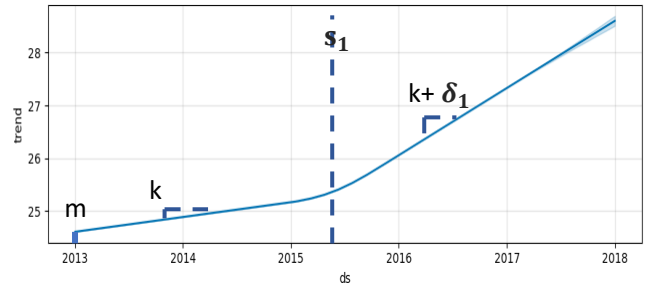
m = offset

$\boldsymbol{\gamma_j}$ = offset to make the function continuous (initialized to $-\boldsymbol{\delta_j s_j}$)

$$a_j(t) = \begin{cases} 1, & \text{if } t \geq s_j, \\ 0, & \text{otherwise.} \end{cases}$$

Logistic (saturated with capacity limit = C)

Trend

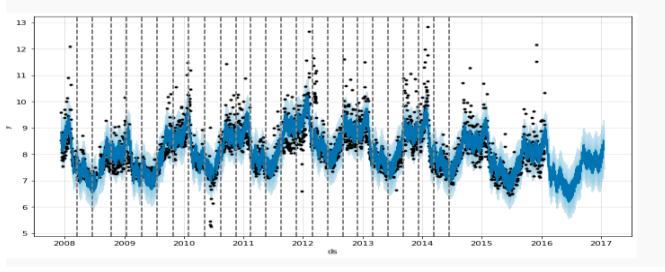


S change points to be defined
[s_1, ...,s_S]
And the variations in the angular coefficient
[δ_1, ...,δ_S]

Prophet Trend

Trend

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$



S

Place the S changepoints

Manually based on domain knowledge, or

Automatically: A high number of potential equidistant changepoints (n_changepoint) are placed in the top 80% of samples (changepoint_range)

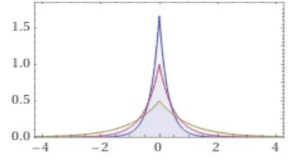
δ

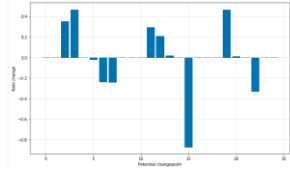
Initialize adjustments

A sparse prior is attributed to the adjustments $\boldsymbol{\delta_j} \sim \boldsymbol{Laplace}(\mathbf{0}, \boldsymbol{\tau})$ with $\boldsymbol{\tau}$ the estimated variance.

If τ is close to zero → hard trend, if τ increases → more non-zero changepoints.

$Laplace(0, \tau)$





$\tau = 0.3$ $\tau = 0.5$ $\tau = 1$

...(fit)...

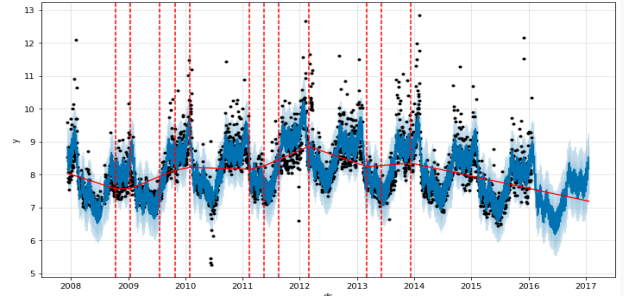
!

Generating future trends

It is assumed that in the future there are the same number of changepoints as a percentage of the samples, so **S** changepoints every **T**
Changepoints will have adjustments distributed as **Laplace(0,"λ")**

$$\lambda = \frac{1}{S}\sum_{j=1}^{S} |\delta_j|$$

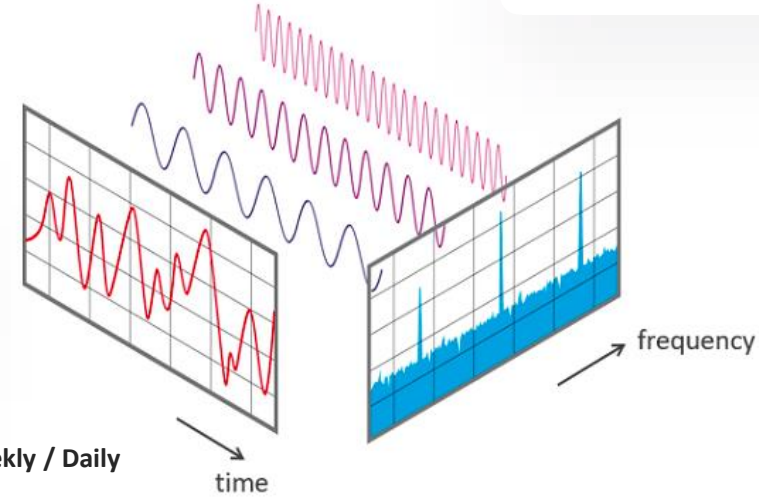
n_changepoint, changepoint_range, τ
tunable hyperparameter



Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

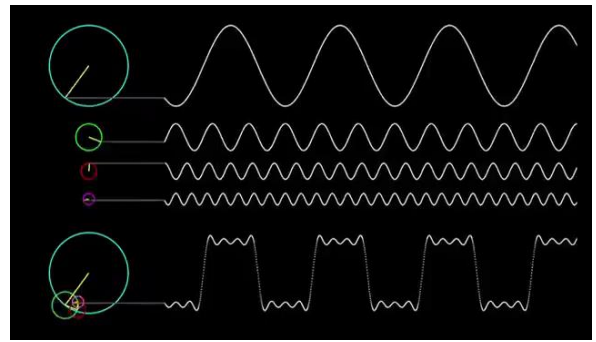


Yearly / Monthly / Weekly / Daily

Teorema (Gauss 1805): A continuous and periodic function can be expressed as the sum of N sinusoidal terms

Nice visualization

https://youtu.be/ds0cmAV-Yek
https://www.youtube.com/watch?v=r6sGWTCMz2k&t=432s



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Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Seasonality

Fourier series sum (of order N → 2N parameters: β ⃗)

$$s(t) = \sum_{n=1}^{N} \left(a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right)$$

$$X(t) = \left[\cos\left(\frac{2\pi(1)t}{P}\right), \ldots, \sin\left(\frac{2\pi(N)t}{P}\right) \right].$$

$$s(t) = X(t)\boldsymbol{\beta}. \quad \boldsymbol{\beta} \sim \text{Normal}(0, \sigma^2) \quad \boldsymbol{\beta} = [a_1, b_1, \ldots, a_N, b_N]^\intercal.$$

With σ^2 the estimated variance of the terms.

N and σ^2 tunable hyperparameter

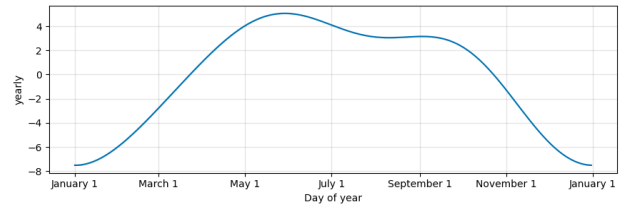
Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

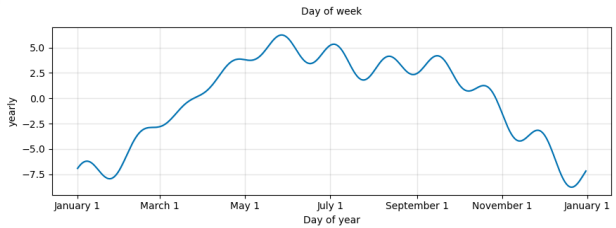
seasonality_prior_scale = 1

Foureier order = 10



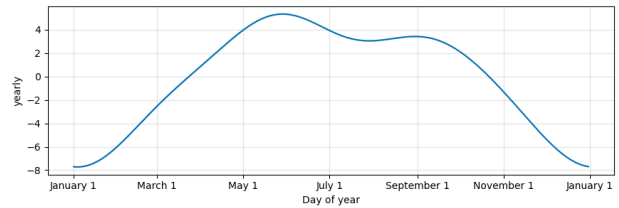
seasonality_prior_scale = 10

Foureier order = 10



seasonality_prior_scale = 10

Foureier order = 2



Prophet Seasonality

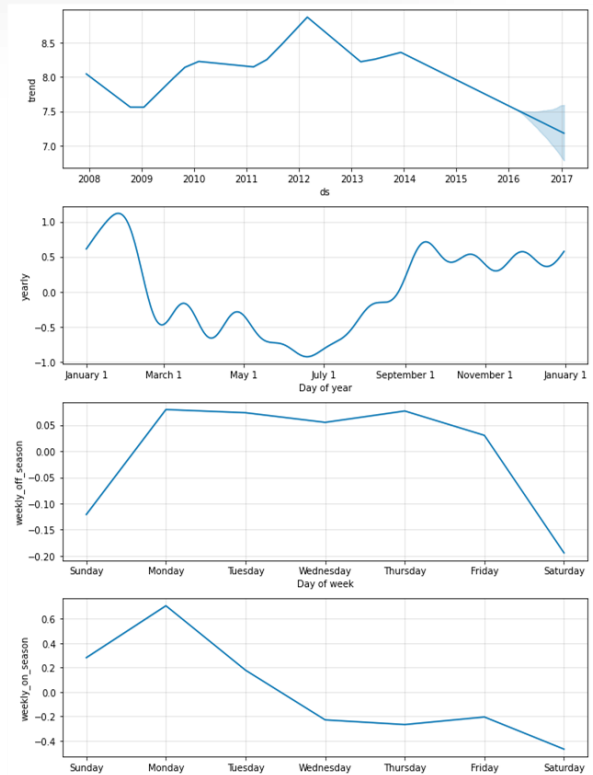
Stagionalità

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

You can use custom seasonalities to differentiate effects based on external features

```python
# Python
m = Prophet(weekly_seasonality=False)
m.add_seasonality(name='weekly_on_season', period=7, fourier_order=3, condition_name='on_season')
m.add_seasonality(name='weekly_off_season', period=7, fourier_order=3, condition_name='off_season')

future['on_season'] = future['ds'].apply(is_nfl_season)
future['off_season'] = ~future['ds'].apply(is_nfl_season)
forecast = m.fit(df).predict(future)
fig = m.plot_components(forecast)
```

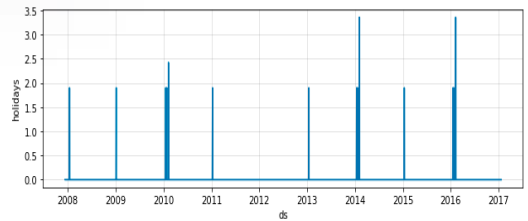


Prophet Events

Events

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Output



Special Events

The effects of the added events are assumed to be independent of each other

Automatically added for many countries
Or
Manually added (0-1 dummies on dates)

You can include upper/lower windows to lengthen the effects to neighboring dates (e.g. Christmas)

Notebooks links

Github Repo

01_data_exploration_plant1.ipynb

02_data_exploration_plant2.ipynb

03_data_exploration_plant2_singles.ipynb

04_data_modelling_plant2_forecasting.ipynb

Q & A

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